

21 Let the e.m.f. of the power supply be E , and the resistance of each lamp be R .

Before filament of lamp K breaks:

$$\text{p.d. across each lamp} = \frac{E}{2}$$

$$\text{Power dissipated in each lamp} = \frac{(E/2)^2}{R} = \frac{E^2}{4R}$$

After filament of lamp K breaks:

$$\text{p.d. across lamp M} = \frac{2E}{3}$$

$$\text{p.d. across lamp L and N} = \frac{E}{3}$$

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$$\text{Power dissipated in lamp M} = \frac{(2E/3)^2}{R} = \frac{4E^2}{9R}$$

$$\text{Power dissipated in lamps L and N} = \frac{(E/3)^2}{R} = \frac{E^2}{9R}$$

Therefore lamp M has become brighter, and lamps L and N has become dimmer.

Answer: C

22 Compare left branch and centre branch: Reading $A = 2(0.6) = 1.2 \text{ A}$